CHEMISTRY

A Comprehensive Textbook for Senior Secondary

Covering: Atomic Structure · Chemical Bonding · States of Matter · Thermodynamics · Equilibrium · Redox Reactions · Organic Chemistry · Electrochemistry · Solutions

National Council of Educational Research and Training — CBSE Pattern

Chapter 1: Structure of the Atom

1.1 Discovery of Sub-Atomic Particles

The atom was once considered indivisible, but experiments in the late 19th and early 20th centuries revealed that atoms are composed of smaller particles: electrons, protons, and neutrons. J. J. Thomson discovered the electron in 1897 using cathode ray tubes. He measured the charge-to-mass ratio (e/m) of the electron as 1.758820×10^{11} C kg^{-1}. Robert Millikan's oil-drop experiment (1909) determined the charge of the electron as 1.6022×10^{-19} C, yielding a mass of 9.1094×10^{-31} kg.

Ernest Rutherford discovered the proton through his gold-foil experiment (1911). Alpha particles were directed at a thin gold foil: most passed through, some deflected, and a very few bounced back. This led Rutherford to propose the nuclear model of the atom — a tiny, dense, positively charged nucleus surrounded by orbiting electrons. The proton has a mass of 1.6726×10^{-27} kg and a charge of $+1.6022 \times 10^{-19}$ C.

James Chadwick discovered the neutron in 1932 by bombarding beryllium with alpha particles. Neutrons are electrically neutral with a mass of 1.6749×10^{-27} kg, slightly greater than that of the proton.

1.2 Atomic Models

Thomson's Plum Pudding Model (1904): Electrons are embedded in a uniform sphere of positive charge, like plums in a pudding. This model could not explain the results of the gold-foil experiment.

Rutherford's Nuclear Model (1911): The atom consists of a very small, dense nucleus containing protons (and later neutrons), with electrons revolving around the nucleus in circular orbits. Limitations: could not explain the stability of atoms or discrete line spectra of elements.

Bohr's Model (1913): Niels Bohr proposed that electrons move in fixed circular orbits (called energy levels or shells, designated n = 1, 2, 3, …) around the nucleus without radiating energy. Energy is emitted or absorbed only when an electron jumps between orbits. The energy of an electron in the n-th orbit of hydrogen is:

$$E_n = -13.6 / n^2 \text{ eV}$$

Bohr's model successfully explained the hydrogen spectrum but failed for multi-electron atoms and could not account for the Zeeman and Stark effects.

1.3 Quantum Mechanical Model

The modern quantum mechanical model treats the electron as a wave-particle. The Heisenberg Uncertainty Principle states that it is impossible to simultaneously determine both the exact position and momentum of an electron: $\Delta x \cdot \Delta p \geq \blacksquare/2$. The Schrödinger wave equation describes

the probability of finding an electron in a region of space, defining atomic orbitals characterized by quantum numbers.

Quantum Numbers:

• Principal quantum number (n): determines the energy level and size of the orbital (n = 1, 2, 3, …).

• Angular momentum quantum number (l): determines the shape (l = 0 to n–1; s, p, d, f).

• Magnetic quantum number (m_l): determines the orientation (m_l = –l to +l).

• Spin quantum number (m_s): +½ or –½.

1.4 Electron Configuration Rules

Aufbau Principle: Electrons fill orbitals in order of increasing energy: 1s → 2s → 2p → 3s → 3p → 4s → 3d → 4p → 5s → 4d → …

Pauli Exclusion Principle: No two electrons in an atom can have the same set of four quantum numbers. Each orbital holds a maximum of 2 electrons with opposite spins.

Hund's Rule: Electrons occupy degenerate orbitals singly with parallel spins before pairing up.

Example configurations:

Carbon (Z=6): $1s^2\ 2s^2\ 2p^2$

Iron (Z=26): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

Copper (Z=29): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$ (anomalous — half-filled d gives extra stability)

Chapter 2: Chemical Bonding and Molecular Structure

2.1 Why Do Atoms Bond?

Atoms bond to achieve a stable electron configuration, usually resembling the nearest noble gas (the octet rule). The driving force is the lowering of potential energy when atoms share, transfer, or pool electrons. Three primary types of chemical bonds exist: ionic, covalent, and metallic.

2.2 Ionic Bonding

Ionic bonds form when one or more electrons are transferred from a metal atom to a non-metal atom. The resulting cation and anion are held together by electrostatic attraction. Example: Sodium chloride (NaCl). Sodium (Na) loses one electron to form Na■, and chlorine (Cl) gains one electron to form Cl■. The lattice energy of NaCl is 786 kJ/mol.

Properties of ionic compounds: high melting and boiling points, conduct electricity when molten or dissolved in water, form crystalline solids, and are generally soluble in polar solvents.

2.3 Covalent Bonding

Covalent bonds form by sharing of electron pairs between atoms. A single covalent bond involves one shared pair (e.g., H■, Cl■), a double bond involves two shared pairs (e.g., O■, C■H■), and a triple bond involves three shared pairs (e.g., N■, C■H■).

Lewis Dot Structures: Represent bonding and lone pairs. For example, water (H■O) has two bonding pairs and two lone pairs on oxygen.

Bond Parameters:

• Bond length: the equilibrium distance between nuclei of two bonded atoms.
• Bond energy: the energy required to break one mole of bonds in gaseous molecules.
• Bond order = (bonding electrons – antibonding electrons) / 2.
• Bond angle: the angle between two adjacent bonds at an atom.

2.4 VSEPR Theory

Valence Shell Electron Pair Repulsion (VSEPR) theory predicts molecular geometry by minimizing repulsion between electron pairs around the central atom. Lone pairs exert greater repulsion than bonding pairs.

Total e■ pairs	Bonding pairs	Lone pairs	Shape	Example	Bond Angle
2	2	0	Linear	BeCl■	180°
3	3	0	Trigonal planar	BF■	120°
3	2	1	Bent (V-shape)	SO■	~119°
4	4	0	Tetrahedral	CH■	109.5°

4	3	1	Trigonal pyramidal	NH■	107°
4	2	2	Bent	H■O	104.5°
5	5	0	Trigonal bipyramidal	PCl■	90°, 120°
6	6	0	Octahedral	SF■	90°

2.5 Hybridization

Hybridization is the mixing of atomic orbitals to form new hybrid orbitals suitable for bonding. Common types:

- **sp hybridization:** linear geometry, 180° bond angle (e.g., BeCl■, C■H■).
- **sp² hybridization:** trigonal planar, 120° bond angle (e.g., BF■, C■H■).
- **sp³ hybridization:** tetrahedral, 109.5° bond angle (e.g., CH■, NH■, H■O).
- **sp³d hybridization:** trigonal bipyramidal (e.g., PCl■).
- **sp³d² hybridization:** octahedral (e.g., SF■).

2.6 Molecular Orbital Theory

In Molecular Orbital (MO) theory, atomic orbitals combine to form molecular orbitals that are delocalized over the entire molecule. Bonding MOs are lower in energy, antibonding MOs are higher. Bond order = ½(bonding electrons – antibonding electrons).

For O■: bond order = ½(10 – 6) = 2 (double bond). O■ is paramagnetic because it has two unpaired electrons in its π* antibonding orbitals — a fact that Lewis structures cannot explain.

Chapter 3: States of Matter — Gases and Liquids

3.1 The Gas Laws

Boyle's Law: At constant temperature, the pressure of a fixed amount of gas is inversely proportional to its volume: PV = constant (P■V■ = P■V■).

Charles's Law: At constant pressure, the volume of a fixed amount of gas is directly proportional to its absolute temperature: V/T = constant (V■/T■ = V■/T■).

Gay-Lussac's Law: At constant volume, the pressure is directly proportional to temperature: P/T = constant.

Avogadro's Law: Equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules: V ∝ n.

3.2 Ideal Gas Equation

$$PV = nRT$$

Where P = pressure (Pa), V = volume (m^3), n = number of moles, R = 8.314 J mol^{-1} K^{-1} (universal gas constant), and T = absolute temperature (K). At STP (273.15 K, 1 atm), one mole of an ideal gas occupies 22.414 L.

3.3 Kinetic Molecular Theory of Gases

Postulates: (1) Gas molecules are in constant, random motion. (2) The volume of individual molecules is negligible compared to the container volume. (3) Intermolecular forces are negligible. (4) Collisions are perfectly elastic. (5) The average kinetic energy is proportional to the absolute temperature: KE = (3/2)kT per molecule, where k = 1.381 × 10^{-23} J/K.

Root mean square speed: $u_{rms} = \sqrt{(3RT/M)}$, where M is the molar mass in kg/mol. For N_2 at 300 K: $u_{rms} \approx 517$ m/s.

3.4 Real Gases — van der Waals Equation

$$(P + an^2/V^2)(V - nb) = nRT$$

Where 'a' corrects for intermolecular attractions and 'b' corrects for the finite volume of molecules. Real gases deviate from ideal behavior at high pressures and low temperatures. The compressibility factor Z = PV/nRT; for an ideal gas Z = 1.

3.5 Liquefaction and Critical Constants

Above the critical temperature (T_c), a gas cannot be liquefied regardless of pressure. The critical pressure (P_c) is the minimum pressure needed to liquefy the gas at T_c. For CO■: T_c = 304.1 K, P_c = 73.9 atm.

Chapter 4: Chemical Thermodynamics

4.1 First Law of Thermodynamics

Energy can neither be created nor destroyed; it can only be transferred or converted from one form to another. Mathematically: $\Delta U = q + w$, where ΔU is the change in internal energy, q is heat absorbed by the system, and w is work done on the system.

For expansion work against constant external pressure: $w = -P_{ext}\Delta V$. At constant volume: $\Delta U = q_v$. At constant pressure: $\Delta H = q_p$, where $H = U + PV$ is the enthalpy.

4.2 Enthalpy Changes

Standard enthalpy of formation (ΔH°_f): The enthalpy change when one mole of a compound is formed from its constituent elements in their standard states.
Examples: $\Delta H^\circ_f(H\blacksquare O, l) = -285.8$ kJ/mol; $\Delta H^\circ_f(CO\blacksquare, g) = -393.5$ kJ/mol.

Hess's Law: The total enthalpy change for a reaction is independent of the pathway: $\Delta H^\circ_{rxn} = \Sigma \Delta H^\circ_f(\text{products}) - \Sigma \Delta H^\circ_f(\text{reactants})$.

4.3 Entropy and the Second Law

Entropy (S) is a measure of the randomness or disorder of a system. The second law states that the total entropy of an isolated system always increases for spontaneous processes: $\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings} > 0$.

Standard molar entropy values increase: solid < liquid < gas. For example, $S^\circ(H\blacksquare O, s) = 48$ J/(mol·K), $S^\circ(H\blacksquare O, l) = 70$ J/(mol·K), $S^\circ(H\blacksquare O, g) = 189$ J/(mol·K).

4.4 Gibbs Free Energy

$$\Delta G = \Delta H - T\Delta S$$

A reaction is spontaneous when $\Delta G < 0$. At equilibrium, $\Delta G = 0$.
- $\Delta H < 0$, $\Delta S > 0 \rightarrow$ spontaneous at all temperatures.
- $\Delta H > 0$, $\Delta S < 0 \rightarrow$ non-spontaneous at all temperatures.
- $\Delta H < 0$, $\Delta S < 0 \rightarrow$ spontaneous at low temperatures.
- $\Delta H > 0$, $\Delta S > 0 \rightarrow$ spontaneous at high temperatures.

Standard free energy: $\Delta G^\circ = -RT \ln K$, connecting thermodynamics to equilibrium.

Chapter 5: Chemical Equilibrium

5.1 Equilibrium in Chemical Reactions

A chemical equilibrium is reached when the rate of the forward reaction equals the rate of the reverse reaction, and the concentrations of reactants and products remain constant over time. Equilibrium is dynamic — reactions continue to occur in both directions.

5.2 The Equilibrium Constant

For the general reaction aA + bB ■ cC + dD:
$K_c = [C]^c[D]^d / [A]^a[B]^b$

$K_p = K_c(RT)^{\Delta n}$, where $\Delta n = (c + d) - (a + b)$ for gaseous reactions. A large K indicates the equilibrium lies far to the right (products favored).

5.3 Le Chatelier's Principle

If a system at equilibrium is subjected to a change in concentration, temperature, or pressure, the system will shift to partially counteract the change and restore equilibrium.

• **Concentration:** Adding reactant shifts equilibrium toward products.
• **Pressure:** Increasing pressure shifts equilibrium toward the side with fewer moles of gas.
• **Temperature:** For exothermic reactions, increasing temperature shifts equilibrium toward reactants; for endothermic reactions, it shifts toward products.

5.4 Ionic Equilibrium — Acids and Bases

Arrhenius: Acids produce H■ in water; bases produce OH■.
Brønsted-Lowry: Acids are proton donors; bases are proton acceptors.
Lewis: Acids are electron-pair acceptors; bases are electron-pair donors.

The pH scale: pH = –log[H■]. At 25°C, pure water has [H■] = [OH■] = 10■■ M, so pH = 7. K_w = [H■][OH■] = 10■¹■ at 25°C.

Strong acids (HCl, H■SO■, HNO■) dissociate completely. Weak acids (CH■COOH, HF) partially dissociate, characterized by K_a. For acetic acid: $K_a = 1.8 \times 10$■■.

5.5 Buffer Solutions

A buffer solution resists changes in pH upon addition of small amounts of acid or base. An acidic buffer contains a weak acid and its conjugate base (e.g., CH■COOH/CH■COO■). The

Henderson-Hasselbalch equation:

$$pH = pK_a + \log([A■]/[HA])$$

Chapter 6: Redox Reactions and Electrochemistry

6.1 Oxidation and Reduction

Oxidation is the loss of electrons (increase in oxidation state); reduction is the gain of electrons (decrease in oxidation state). They always occur simultaneously. The substance that is oxidized is the reducing agent; the substance that is reduced is the oxidizing agent.

Rules for assigning oxidation states:
• Free elements: 0
• Monatomic ions: equal to the ion charge
• Oxygen: –2 (except in peroxides: –1, superoxides: –½, and OF■: +2)
• Hydrogen: +1 (except in metal hydrides: –1)
• Fluorine: always –1
• Sum of oxidation states = charge on the species

6.2 Balancing Redox Reactions

Half-reaction method: (1) Split into oxidation and reduction half-reactions. (2) Balance atoms other than O and H. (3) Balance O by adding H■O. (4) Balance H by adding H■ (acidic) or OH■ (basic). (5) Balance charge by adding electrons. (6) Equalize electrons and add the two half-reactions.

Example in acidic solution:
MnO■■ + 8H■ + 5e■ → Mn²■ + 4H■O (reduction)
Fe²■ → Fe³■ + e■ (oxidation) × 5
Overall: MnO■■ + 8H■ + 5Fe²■ → Mn²■ + 5Fe³■ + 4H■O

6.3 Electrochemical Cells

A galvanic (voltaic) cell converts chemical energy to electrical energy. The Daniel cell: Zn(s) | Zn²■(aq) || Cu²■(aq) | Cu(s). Zinc is oxidized at the anode; copper is reduced at the cathode. $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = +0.34 - (-0.76) = +1.10$ V.

6.4 Nernst Equation

$$E_{cell} = E^\circ_{cell} - (RT/nF) \ln Q$$

At 25°C: $E_{cell} = E^\circ_{cell} - (0.0592/n) \log Q$. At equilibrium, $E_{cell} = 0$ and $Q = K$, so $E^\circ_{cell} = (0.0592/n) \log K$. $\Delta G^\circ = -nFE^\circ_{cell}$.

6.5 Electrolysis and Faraday's Laws

First Law: The mass of substance deposited at an electrode is proportional to the quantity of electricity passed: $m = (M \times I \times t) / (n \times F)$, where $F = 96485$ C/mol.
Second Law: When the same quantity of electricity is passed through different electrolytes, the masses deposited are proportional to their equivalent masses.

Chapter 7: Organic Chemistry — Some Basic Principles

7.1 Classification of Organic Compounds

Organic compounds are classified based on their carbon skeleton and functional groups. The main classes are:

• **Alkanes (C_nH_{2n+2}):** saturated hydrocarbons, single bonds only. E.g., methane (CH■), ethane (C■H■), propane (C■H■).

• **Alkenes (C_nH_{2n}):** one C=C double bond. E.g., ethene (C■H■).

• **Alkynes (C_nH_{2n-2}):** one C≡C triple bond. E.g., ethyne (C■H■).

• **Aromatic:** contain a benzene ring (C■H■).

7.2 IUPAC Nomenclature

The IUPAC naming system: (1) Find the longest carbon chain (parent chain). (2) Number from the end nearest the first substituent. (3) Name substituents as prefixes. (4) Use the suffix to indicate the functional group: -ane (alkane), -ene (alkene), -yne (alkyne), -ol (alcohol), -al (aldehyde), -one (ketone), -oic acid (carboxylic acid).

Examples:

• CH■CH■CH■CH■ → butane

• CH■CH(CH■)CH■CH■ → 2-methylbutane

• CH■CH■OH → ethanol

• CH■COOH → ethanoic acid (acetic acid)

• CH■COCH■ → propan-2-one (acetone)

7.3 Isomerism

Structural isomerism: same molecular formula, different connectivity. Types include chain, position, and functional group isomerism.

Stereoisomerism: same connectivity, different spatial arrangement. Includes geometrical (cis-trans) isomerism and optical isomerism (chirality).

Example: C■H■■ has two structural isomers — n-butane and isobutane (2-methylpropane). C■H■Cl■ (1,2-dichloroethene) has cis and trans geometrical isomers.

7.4 Reaction Mechanisms

Organic reactions proceed through intermediates. Three main types of bond breaking:

• **Homolytic fission:** each atom gets one electron → free radicals → radical reactions.

• **Heterolytic fission:** both electrons go to one atom → carbocation/carbanion → ionic reactions.

Types of reagents: nucleophiles (electron-pair donors, e.g., OH■, CN■, NH■) and electrophiles (electron-pair acceptors, e.g., H■, NO■■, AlCl■).

Inductive Effect: Electron-withdrawing groups (–NO■, –CN, –COOH, halogens) pull electron density along sigma bonds (–I effect). Electron-donating groups (alkyl groups) push electron density (+I effect). This affects acidity, basicity, and reactivity.

Chapter 8: Solutions and Colligative Properties

8.1 Types of Solutions and Concentration

A solution is a homogeneous mixture of two or more substances. The substance present in larger amount is the solvent; the other is the solute. Concentration can be expressed as:
• **Molarity (M):** moles of solute per liter of solution.
• **Molality (m):** moles of solute per kilogram of solvent.
• **Mole fraction (x):** moles of component / total moles.
• **Mass percentage:** (mass of solute / mass of solution) × 100.

8.2 Raoult's Law

For an ideal solution, the partial vapor pressure of each volatile component is equal to the product of its mole fraction and its pure vapor pressure: $P_A = x_A \cdot P°_A$. Positive deviations occur when solute-solvent interactions are weaker than pure component interactions (e.g., ethanol-water). Negative deviations occur when interactions are stronger (e.g., HCl-water).

8.3 Colligative Properties

Properties that depend on the number of solute particles, not their identity:

1. Relative lowering of vapor pressure: $\Delta P/P° = x_{solute}$

2. Elevation of boiling point: $\Delta T_b = K_b \cdot m$, where K_b for water = 0.512 K·kg/mol.

3. Depression of freezing point: $\Delta T_f = K_f \cdot m$, where K_f for water = 1.86 K·kg/mol.

4. Osmotic pressure: $\pi = iMRT$, where i = van't Hoff factor (accounts for dissociation or association). For NaCl (complete dissociation): i = 2.

8.4 Abnormal Molar Masses

Electrolytes in solution dissociate, producing more particles than expected, leading to higher observed colligative properties and lower calculated molar mass (abnormal). The van't Hoff factor corrects this: i = (observed property) / (calculated property). For benzoic acid in benzene, i < 1 because of association (dimerization).

Practice Numerical Problems

1. Calculate the wavelength of an electron moving with a velocity of 2.05 × 10■ m/s (mass of electron = 9.1 × 10■^{31} kg, h = 6.626 × 10■^{3}■ J·s). [Ans: 3.55 × 10■^{1}■ m]

2. A sample of gas occupies 500 mL at 27°C and 1 atm. What volume will it occupy at STP? [Ans: 455.4 mL]

3. Calculate the enthalpy of combustion of methane given: ΔH°f(CO■) = –393.5 kJ/mol, ΔH°f(H■O, l) = –285.8 kJ/mol, ΔH°f(CH■) = –74.8 kJ/mol. [Ans: –890.3 kJ/mol]

4. The pH of a 0.10 M solution of acetic acid (Ka = 1.8 × 10■■) is? [Ans: pH ≈ 2.87]

5. Calculate the EMF of the cell: Zn | Zn^{2}■(0.001 M) || Ag■(0.1 M) | Ag. Given E°(Zn^{2}■/Zn) = –0.76 V, E°(Ag■/Ag) = +0.80 V. [Ans: +1.503 V]

6. The freezing point depression of a 0.5 m aqueous NaCl solution is? (Kf = 1.86 K·kg/mol, i = 1.85). [Ans: 1.72°C]

7. Name the following compound: CH■CH(OH)CH■CH■. [Ans: butan-2-ol]

8. What mass of copper is deposited when a current of 2 A is passed through CuSO■ solution for 30 minutes? (Cu = 63.5, F = 96485 C/mol). [Ans: 1.18 g]

9. Calculate the bond order of N■ and N■■ using MO theory. [Ans: N■ = 3, N■■ = 2.5]

10. A solution contains 18 g of glucose (C■H■■O■, M = 180) in 500 g of water. Calculate its osmotic pressure at 300 K. (R = 0.0821 L·atm/mol·K). [Ans: 4.93 atm]